Drone Payload Team 3 Teamwork Agreement

For the MathWorks Classroom Challenge Projects

Project: Drone Payload Capacity and Structural Design Analysis

Team Name: Drone Payload Team 3 (DPT3)

Program/Class: Engineering Pathways Program/MathWorks Challenge

Members and Roles

1. Julianna Amaya - Quality Assurance & Reproducibility Lead, Modeling Lead
2. Boula (Paul) Etans - Modeling Lead, Documentation & Visualization Lead
3. Mario Mendoza - Analysis/Validation Lead
4. Nathan Castro - Project Manager

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs

- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs code to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Discord
- Secondary channel: Messages
- Response time: within 24 hours on weekdays and within 48 hours on weekends
- Meeting cadence: Meetings, Mondays, 8 pm, 60-90 minutes (Via. Discord)
- Decision records: Google Docs located in a Google Drive

4) Collaboration & Tools

- Version control: One shared GitHub and Google Drive
- Google Drive and README.md

File naming & style: descriptive names, *three_different_words*

- Backups: PM tags old versions according to numbered system

5) Quality Standards (Definition of 'Done')

Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report

MathWorks Internship

June 30th	**Internship Orientation**
July 6 - August 5th	**Main phase:** Teams work on the projects and MathWorks provides Office Hours
August 6th - August 7th	**Wrapup phase**: Teams complete their reports, and online submissions
August 10th - August 14th	**Evaluation:** First round of peer reviews organized by EPP to provide five nominations. Followed by EPP+MathWorks selecting 3 winners.

6) Timeline & Milestones

- Week 1: Complete Task 1 (Gather your starter assumptions and load material properties) found in “DroneDesign_StudentProjectTemplate_Preview.pdf” located in MathWorks Project Repository. Sketch 2 designs that’ll be used for the project.

- Week 2: Finish task 2 (Propose at least two drone arm designs) by utilizing CAD to create 3D models of the sketched designs, including dimensions and faces. Finish task 3 (Perform thrust-to-weight analysis across all designs and material options) by inputting both designs to pass the project requirements

- Week 3: Finish Task 4 (Perform finite element analysis across all design and material options) with both designs implementing the required six materials. Determine the final recommendation for drone design and materials that includes best overall structural integrity and performance and meets project requirements.

- Week 4: Complete Task 5 (Optimize the total material cost of the drone arm) by gathering the best low-cost material to use for both designs. Document the project and input it into a GitHub Repository.

7) Decision-Making & Conflict Resolution

- Default: consensus after discussion; PM facilitates
- If blocked >48 hrs: Time-limited debate, majority vote, PM documents decision
- Technical disagreements: Evidence-based comparison before vote
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: To-Do / Doing / Review / Done with owners and due dates documented in README.md

- Accountability: Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution

- Attendance: If you're late to a meeting, give a heads-up | If you'll miss a meeting, 24-hour notice and share updates asynchronously.
- We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- [ ] Develop at least two distinct drone arm designs by drawing
- [ ] At least two drone design sketches showing key geometric features
- [ ] Results of thrust-to-weight analysis presented either in a summary table or plot,

  Results of finite element analysis
- [ ] Numerical values presented in a summary table
- [ ] Visualizations of x-, y-, and z-displacement and Von Mises stress
- [ ] A final design recommendation justified using the results of your analyses
- [ ] Code repository with clear structure and README
- [ ] Hand calculations linked to modeled results for cross-validation
- [ ] Specific figures & visualizations

- Final report with specified sections

12) Agreement & Signatures

Name & Signature: Nathan Castro Date: July 6, 2026

Name & Signature: Boula Etans Date: July 6, 2026

Name & Signature: Mario Mendoza Jr. Date: July 6, 2026

Name & Signature: Julianna Amaya Date: July 6, 2026